Asset Pricing Assignment 1

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Outputs

1. Regression table for the panel data regressions

Table 1: Panel Regression Results

	(1) Pooled OLS	(2) Fama–MacBeth	(3) Two-way FE	(4) Two-way FE + Cluster
β_{MKT}	0.0023***	0.0033	-0.0002	-0.0002
	(0.0002)	(0.0035)	(0.0003)	(0.0019)
β_{SMB}	-0.0006***	-0.0009	-0.0000	-0.0000
	(0.0001)	(0.0017)	(0.0002)	(0.0013)
β_{HML}	0.0028***	0.0060**	0.0026***	0.0026
	(0.0002)	(0.0025)	(0.0002)	(0.0016)
β_{RMW}	0.0014***	0.0023	0.0005***	0.0005
	(0.0001)	(0.0019)	(0.0001)	(0.0009)
β_{CMA}	-0.0017***	-0.0007	-0.0001	-0.0001
	(0.0001)	(0.0013)	(0.0001)	(0.0010)
β_{MOM}	0.0001	-0.0008	-0.0030***	-0.0030*
	(0.0002)	(0.0027)	(0.0002)	(0.0016)
β_{CAT}	-0.0004***	-0.0007*	0.0000	0.0000
	(0.0001)	(0.0004)	(0.0001)	(0.0004)
Observations	630,807	89	630,807	630,807
R^2	0.0024	–	0.0007	0.0007

Notes: Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.
Notes: For Fama-MacBeth, observations refer to the number of monthly cross-sectional regressions.

2. Summary statistics for the CAT factor

Table 2: Summary Statistics for the CAT Factor

	Mean	Std. Dev.	MKT	SMB	HML	RMW	CMA	MOM
CAT	0.0005	0.0091	0.4094	0.3481	0.0425	-0.2154	-0.0780	-0.2845

Notes: MKT denotes $R_m - R_f$. The remaining columns report correlations with CAT.

The summary statistics for the CAT factor serve as a preliminary check of whether the factor is economically meaningful. In particular, the mean and standard deviation indicate whether the factor has a non-trivial return and sufficient variation, while the correlations with other factors help assess whether it captures distinct information beyond existing factors.

Discussion

3. Validation of the Seven-Factor Model

The regression results provide limited support for the seven-factor model. While several factors, such as MKT and RMW, appear significant in the pooled OLS specification, their significance largely disappears in the Fama–MacBeth regressions. Furthermore, in the two-way fixed effects specifications, most coefficients are close to zero and the R^2 is extremely low (around 0.0007), indicating that the model has little explanatory power once firm and time effects are controlled for. The CAT factor, in particular, does not show robust significance across specifications, suggesting that it does not represent a distinct priced risk factor.

4. Comparison Across Specifications

The differences across specifications arise from how each method handles variation in the data. Pooled OLS combines time-series and cross-sectional variation and tends to overstate significance. In contrast, the Fama–MacBeth approach isolates cross-sectional variation, leading to larger standard errors and weaker significance. The two-way fixed effects models further remove firm and time heterogeneity, causing coefficients to shrink toward zero. Finally, clustering standard errors increases them further, reinforcing that the apparent significance in pooled OLS is not robust.

5. GRS test results

Table 3: GRS Test Results

Model	GRS F-statistic	p-value	Avg. $\lvert\alpha\rvert$
Seven-factor model	9.1054	0.0000	0.0037
Six-factor model	9.0167	0.0000	0.0037

Notes: The GRS test evaluates the joint null hypothesis that all 25 portfolio intercepts are zero.

The seven-factor model does not improve the GRS performance relative to the six-factor model; the GRS statistic slightly increases from 9.0167 to 9.1054, while the average absolute alpha remains essentially unchanged at 0.0037.

The GRS test results provide a joint test of whether the model fully explains the cross-section of returns. The very small p-values indicate that the null hypothesis is rejected for both models, suggesting that neither the six-factor nor the seven-factor model adequately prices the 25 portfolios. Moreover, the similarity in the GRS statistics and average absolute alphas suggests that adding the CAT factor does not meaningfully improve model performance.

6. Alpha Heatmaps

The alpha heatmaps suggest that there are some patterns in the pricing errors, rather than a completely random distribution around zero. Rather than being randomly distributed around zero, most of the alphas are negative, as indicated by the dominance of blue shades across both heatmaps. This suggests that the model tends to systematically overpredict expected returns for a large number of portfolios.

The largest pricing errors are concentrated in the corner portfolios, particularly those with extreme CAT exposure. In the bottom-left and top-left regions of the heatmap (low HML portfolios combined with extreme CAT quintiles), the alphas are strongly negative and visually stand out as darker blue areas. This indicates that the model performs poorly for portfolios with extreme CAT loadings.

In contrast, the middle portfolios—especially those around the center of the grid—show alphas closer to zero, as reflected by lighter colors. This suggests that the model fits relatively well for firms with more moderate characteristics.

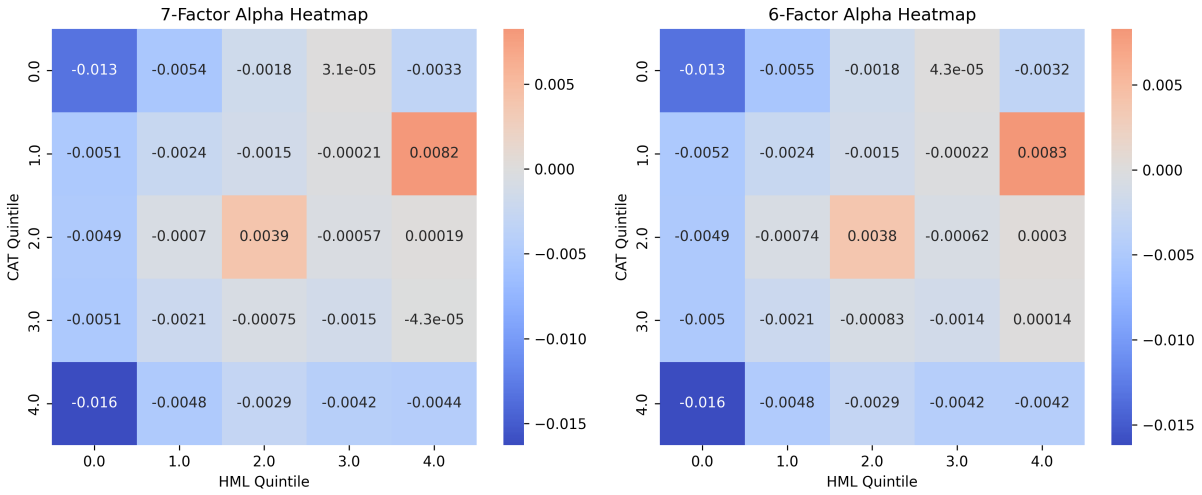

Figure 1: Alpha Heatmaps for 5×5 Portfolios

There are also notable exceptions. For instance, in the region corresponding to moderate CAT exposure and high HML loading, the heatmap shows a distinct red cell, indicating a positive alpha. This implies that the model underpredicts returns for that specific group of portfolios.

Comparing the six-factor and seven-factor heatmaps, the overall pattern remains very similar. The addition of the CAT factor does not meaningfully reduce the magnitude or the structure of the pricing errors. The persistence of strong negative alphas in extreme portfolios suggests that the seven-factor model still fails to capture important nonlinearities or interaction effects between CAT and HML.
Overall, the heatmaps provide visual evidence that the model does not adequately price the 25 portfolios, and that the inclusion of the CAT factor does not substantially improve model performance.